1

This functions quite similar to a logistic regression

So assumptions are:

- $0 \leq C_i(x) \leq 1$ and is fixed
- n samples with $\forall y^{(i)} \in y, y^{(i)} \in \{0, 1\}$
- bias fixed at 0.5
- We will use a loss function of logistic loss
- $C(x^{(i)})$ returns a list of probabilities corresponding such that the jth probability is $C_j(x^{(i)})$ or the probability that the jth model predicts for our input $x^{(i)}$
- $\sigma(z)$ same as logistic regression so $\sigma(z) = 1$ iff $z \geq 0$ else 0
- η and epoch numbers are given to us

Algorithm pseudocode:

```
m = # Models
n = # Data Points

w = [None] * m

for w_j in w:
  w_j = 1/m

p = [None] * n

for p_i in p:
  p_i = C(x^{(i)})

for i in range(NUM_EPOCHS):
  z = w^T p - 0.5
  y_pred = sigma(z)
  dw = dotproduct(x, y_pred-y)
  w -= dw * eta

return w
```

2

We find that for correct classifications, we multiply by $1/\sqrt{3}$ and for incorrect classigications we multiply by $\sqrt{3}$. Thus $e^{-\alpha} = 1/\sqrt{3}$

This means that $\alpha = \ln\sqrt{3}$

We know by the definition of α that:

$$\alpha = 0.5\ln(\tfrac{1-\epsilon}{\epsilon})$$

So we get the following:

$$\ln\sqrt{3} = \alpha = 0.5\ln(\tfrac{1-\epsilon}{\epsilon})$$

$$\rightarrow \ln\sqrt{3} = 0.5\ln(\tfrac{1-\epsilon}{\epsilon})$$

$$\rightarrow 0.5\ln 3 = 0.5\ln(\tfrac{1-\epsilon}{\epsilon})$$

$$\rightarrow 3 = \tfrac{1-\epsilon}{\epsilon}$$

$$\rightarrow 3\epsilon = 1 - \epsilon$$

$$\rightarrow \epsilon = \tfrac{1}{4}$$

3

We are given the initial following features:

- $x_1^{(k)} = k \times 10^{-1}$
- $x_2^{(k)} = k$
- $x_3^{(k)} = k \times 10$

Thus our degree 2 polynomial transformation will result in the following features:

- $x_0^{(k)} = 1$
- $x_1^{(k)} = k \times 10^{-1}$
- $x_2^{(k)} = k$
- $x_3^{(k)} = k \times 10$
- $x_4^{(k)} = (x_1^{(k)})^2 = k^2 \times 10^{-2}$
- $x_5^{(k)} = (x_2^{(k)})^2 = k^2$
- $x_6^{(k)} = (x_3^{(k)})^2 = k^2 \times 10^2$
- $x_7^{(k)} = x_1^{(k)} x_2^{(k)} = k^2 \times 10^{-1}$
- $x_8^{(k)} = x_1^{(k)} x_3^{(k)} = k^2 \times 10^1$
- $x_9^{(k)} = x_2^{(k)} x_3^{(k)} = k^2$

4

Given our initial centroids are $c_1 = (2, 0)$ and $c_2 = (-2, 0)$ with the below initial points:

- $x^{(1)} = (-1, 0)$
- $x^{(2)} = (1.5, 0)$
- $x^{(3)} = (-1, -1)$
- $x^{(4)} = (1, 1)$
- $x^{(5)} = (2, 1)$

After this iteration we find that $x^{(1)}, x^{(3)}$ are closest to c_2 and that $x^{(2)}, x^{(4)}, x^{(5)}$ are closest to c_1

And membership is for cluster 1:

- $x^{(2)}, x^{(4)}, x^{(5)}$

Thus membership is for cluster 2:

- $x^{(1)}, x^{(3)}$

Thus we find our new centroids are:

- $c_1 = ([1.5 + 1 + 2]/3, [0 + 1 + 1]/3) = (1.5, \frac{2}{3})$
- $c_2 = ([-1 - 1]/2, [0 - 1]/2) = (-1, 0.5)$

5

Were assumin Euclidian distances and we have the two below clusters:

Cluster 1:

- $(-1, 1)$
- $(-1, -1)$

Cluster 2:

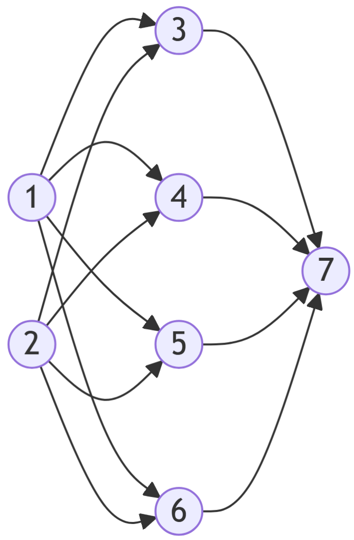
- $(1, 1)$
- $(1, -1)$

The single linkage is the minimum distance between points of the different clusters which in this case is a Euclidian distance of 2 for $(-1, 1)$ in cluster 1 and $(1, 1)$ in cluster 2.

The maximum linkage is the minimum distance between points of the different clusters which in this case is a Euclidian distance of $2\sqrt{2}$ for $(-1, -1)$ in cluster 1 and $(1, 1)$ in cluster 2.

6

a



b

For simplicity, we will fo with weights $w_{13}, w_{14}, w_{15}, w_{16}$

w_{13}

$$\frac{dL_w}{dw_{13}} = \frac{dL_w}{da_7}\frac{da_7}{da_3}\frac{da_3}{dw_{13}}$$

$$\rightarrow \frac{dL_w}{dw_{13}} = (-2(y-a_7))(a_3(1-a_3)w_{37})(a_3(1-a_3)a_1)$$

w_{14}

$$\frac{dL_w}{dw_{14}} = \frac{dL_w}{da_7}\frac{da_7}{da_4}\frac{da_4}{dw_{14}}$$

$$\rightarrow \frac{dL_w}{dw_{14}} = (-2(y-a_7))(a_4(1-a_4)w_{47})(a_4(1-a_4)a_1)$$

w_{15}

$$\frac{dL_w}{dw_{15}} = \frac{dL_w}{da_7}\frac{da_7}{da_5}\frac{da_5}{dw_{15}}$$

$$\rightarrow \frac{dL_w}{dw_{15}} = (-2(y-a_7))(a_5(1-a_5)w_{57})(a_5(1-a_5)a_1)$$

w_{16}

$$\frac{dL_w}{dw_{16}} = \frac{dL_w}{da_7}\frac{da_7}{da_6}\frac{da_6}{dw_{16}}$$

$$\rightarrow \frac{dL_w}{dw_{16}} = (-2(y-a_7))(a_6(1-a_6)w_{67})(a_6(1-a_6)a_1)$$

c

$\sigma(z) = \frac{1}{1+e^{-z}}$

with all weights being 1 and biases being 0 on the input $(0,0)$:

- input is $(0,0)$ so $a_1 = a_2 = 0$
- $a_3 = \sigma(w_{13}a_1 + w_{23}a_2) = \sigma(1*0+1*0) = \frac{1}{2}$
- $a_4 = \sigma(w_{14}a_1 + w_{24}a_2) = \sigma(1*0+1*0) = \frac{1}{2}$
- $a_5 = \sigma(w_{15}a_1 + w_{25}a_2) = \sigma(1*0+1*0) = \frac{1}{2}$
- $a_6 = \sigma(w_{16}a_1 + w_{26}a_2) = \sigma(1*0+1*0) = \frac{1}{2}$

Then determining the output:

- $\hat{y} = a_7 = \sigma(w_{37}a_3 + w_{47}a_4 + w_{57}a_5 + w_{67}a_6) = \sigma(\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+\frac{1}{2}) = \frac{1}{1+e^{-2}} = 0.8808$

d

Given that $y = 1$:

w_{13}

$$\frac{dL_w}{dw_{13}} = \frac{dL_w}{da_7}\frac{da_7}{da_3}\frac{da_3}{dw_{13}}$$

$$\rightarrow \frac{dL_w}{dw_{13}} = (-2(y-a_7))(a_3(1-a_3)w_{37})(a_3(1-a_3)a_1)$$

$$\rightarrow \frac{dL_w}{dw_{13}} = (-2(1-0.8808))(0.5(1-0.5)1)(0.5(1-0.5)0) = 0$$

w_{14}

$$\frac{dL_w}{dw_{14}} = \frac{dL_w}{da_7}\frac{da_7}{da_4}\frac{da_4}{dw_{14}}$$

$$\rightarrow \frac{dL_w}{dw_{14}} = (-2(y-a_7))(a_4(1-a_4)w_{47})(a_4(1-a_4)a_1)$$

$$\rightarrow \frac{dL_w}{dw_{14}} = (-2(1-0.8808))(0.5(1-0.5)1)(0.5(1-0.5)0) = 0$$

w_{15}

$$\frac{dL_w}{dw_{15}} = \frac{dL_w}{da_7}\frac{da_7}{da_5}\frac{da_5}{dw_{15}}$$

$$\rightarrow \frac{dL_w}{dw_{15}} = (-2(y - a_7))(a_5(1 - a_5)w_{57})(a_5(1 - a_5)a_1)$$

$$\rightarrow \frac{dL_w}{dw_{15}} = (-2(1 - 0.8808))(0.5(1 - 0.5)1)(0.5(1 - 0.5)0) = 0$$

w_{16}

$$\frac{dL_w}{dw_{16}} = \frac{dL_w}{da_7}\frac{da_7}{da_6}\frac{da_6}{dw_{16}}$$

$$\rightarrow \frac{dL_w}{dw_{16}} = (-2(y - a_7))(a_6(1 - a_6)w_{67})(a_6(1 - a_6)a_1)$$

$$\rightarrow \frac{dL_w}{dw_{16}} = (-2(1 - 0.8808))(0.5(1 - 0.5)1)(0.5(1 - 0.5)0) = 0$$

They all are 0 because the inputs were originally 0 so multiplying be 0 :)

7

Sample	x_1	x_2	y
0	0.4967	-0.1383	1
1	0.6477	1.5230	1
2	-0.2342	-0.2341	0
3	1.5792	0.7674	1
4	-0.4695	0.5426	1
5	-0.4634	-0.4657	0
6	0.2420	-1.9133	0
7	-1.7249	-0.5623	0
8	-1.0128	0.3142	0
9	-0.9080	-1.4123	0
10	1.4656	-0.2258	1
11	0.0675	-1.4247	0
12	-0.5444	0.1109	0
13	-1.1510	0.3757	0
14	-0.6006	-0.2917	0
15	-0.6017	1.8523	1
16	-0.0135	-1.0577	0
17	0.8225	-1.2208	0
18	0.2089	-1.9597	0
19	-1.3282	0.1969	0

b and c

Set Num	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	**17**	11	1	9	3	**13**	**15**	**14**	**7**	**13**	**7**	**15**	**12**	**17**	**14**	**12**	8	**14**	**12**	0
2	**6**	**8**	**0**	**11**	**7**	10	18	16	**7**	**2**	**2**	**0**	4	9	**6**	**8**	**6**	**8**	**7**	**11**
3	1	**0**	15	**4**	**2**	11	7	**2**	**0**	**2**	**4**	**14**	**13**	**2**	**0**	**4**	**13**	6	8	**14**
4	**14**	**9**	**12**	**18**	**6**	16	19	**3**	4	**6**	**12**	**14**	10	**3**	**12**	**6**	**18**	1	**9**	**12**
5	**5**	**11**	**11**	**19**	10	**6**	**0**	**0**	**19**	12	**8**	2	**6**	**5**	7	**8**	4	**0**	18	9
6	**11**	14	8	**19**	**16**	**16**	**19**	**11**	6	**1**	2	**16**	**4**	**16**	**16**	**16**	**1**	**1**	**4**	0
7	**0**	18	**1**	11	**5**	3	**10**	16	**5**	4	19	**1**	**5**	**10**	**15**	**15**	**0**	8	**5**	**15**
8	**2**	**19**	3	**18**	**2**	**18**	**19**	**6**	**19**	8	**0**	**7**	**6**	**17**	**7**	**0**	10	**17**	9	**2**
9	6	**15**	**15**	**19**	16	1	**0**	**15**	11	**4**	**4**	**8**	**8**	**2**	18	**15**	**15**	**2**	**19**	**0**
10	**19**	10	16	**7**	**3**	5	**7**	**19**	**2**	**15**	**2**	**17**	13	**17**	1	**2**	**15**	8	**3**	0

e

Suppose we have the sample point $(0.5, -0.2)$

We predict for each model what this point is. If the majority say that it is a class we will output that class. In the case of a draw we can in our case default to class 1.

For this point, we find that this is the list of predictions from the different models: [0101111110]

Thus since 7 of the 10 models predicted class 1 the majority vote is for this data point to be predicted as class 1.

8

Sample	x_1	x_2	y
0	0.4967	-0.1383	1
1	0.6477	1.5230	1
2	-0.2342	-0.2341	-1
3	1.5792	0.7674	1
4	-0.4695	0.5426	1
5	-0.4634	-0.4657	-1
6	0.2420	-1.9133	-1
7	-1.7249	-0.5623	-1
8	-1.0128	0.3142	-1
9	-0.9080	-1.4123	-1
10	1.4656	-0.2258	1

Sample	x_1	x_2	y
11	0.0675	-1.4247	-1
12	-0.5444	0.1109	-1
13	-1.1510	0.3757	-1
14	-0.6006	-0.2917	-1
15	-0.6017	1.8523	1
16	-0.0135	-1.0577	-1
17	0.8225	-1.2208	-1
18	0.2089	-1.9597	-1
19	-1.3282	0.1969	-1

b and c and d

Stump #	Feature	Threshold	α_j	Weights
1	1	0.4591	1.0986	‘2.50e-01’, ‘2.78e-02’, ‘2.78e-02’, ‘2.78e-02’, ‘2.78e-02’, ‘2.78e-02’, ‘2.78e-02’, ‘2.78e-02’, ‘2.78e-02’, ‘2.78e-02’, ‘2.50e-01’, ‘2.78e-02’, ‘2.78e-02’, ‘2.78e-02’, ‘2.78e-02’, ‘2.78e-02’, ‘2.78e-02’, ‘2.78e-02’, ‘2.78e-02’, ‘2.78e-02’ –
2	0	0.3693	1.1989	‘1.36e-01’, ‘1.52e-02’, ‘1.52e-02’, ‘1.52e-02’, ‘1.67e-01’, ‘1.52e-02’, ‘1.52e-02’, ‘1.52e-02’, ‘1.52e-02’, ‘1.52e-02’, ‘1.36e-01’, ‘1.52e-02’, ‘1.52e-02’, ‘1.52e-02’, ‘1.52e-02’, ‘1.67e-01’, ‘1.52e-02’, ‘1.67e-01’, ‘1.52e-02’, ‘1.52e-02’ –
3	1	-0.2300	1.3704	‘7.26e-02’, ‘8.06e-03’, ‘8.06e-03’, ‘8.06e-03’, ‘8.87e-02’, ‘8.06e-03’, ‘8.06e-03’, ‘8.06e-03’, ‘1.25e-01’, ‘8.06e-03’, ‘7.26e-02’, ‘8.06e-03’, ‘1.25e-01’, ‘1.25e-01’, ‘8.06e-03’, ‘8.87e-02’, ‘8.06e-03’, ‘8.87e-02’, ‘8.06e-03’, ‘1.25e-01’ –
4	1	0.4591	0.8865	‘2.50e-01’, ‘4.72e-03’, ‘4.72e-03’, ‘4.72e-03’, ‘5.19e-02’, ‘4.72e-03’, ‘4.72e-03’, ‘4.72e-03’, ‘7.31e-02’, ‘4.72e-03’, ‘2.50e-01’, ‘4.72e-03’, ‘7.31e-02’, ‘7.31e-02’, ‘4.72e-03’, ‘5.19e-02’, ‘4.72e-03’, ‘5.19e-02’, ‘4.72e-03’, ‘7.31e-02’ –

Stump #	Feature	Threshold	α_j	Weights
5	0	-0.5069	0.9414	‘1.44e-01’, ‘2.72e-03’, ‘1.79e-02’, ‘2.72e-03’, ‘2.99e-02’, ‘1.79e-02’, ‘1.79e-02’, ‘2.72e-03’, ‘4.21e-02’, ‘2.72e-03’, ‘1.44e-01’, ‘1.79e-02’, ‘4.21e-02’, ‘4.21e-02’, ‘2.72e-03’, ‘1.96e-01’, ‘1.79e-02’, ‘1.96e-01’, ‘1.79e-02’, ‘4.21e-02’ –
6	1	-0.2300	0.7982	‘8.66e-02’, ‘1.63e-03’, ‘1.07e-02’, ‘1.63e-03’, ‘1.80e-02’, ‘1.07e-02’, ‘1.07e-02’, ‘1.63e-03’, ‘1.25e-01’, ‘1.63e-03’, ‘8.66e-02’, ‘1.07e-02’, ‘1.25e-01’, ‘1.25e-01’, ‘1.63e-03’, ‘1.18e-01’, ‘1.07e-02’, ‘1.18e-01’, ‘1.07e-02’, ‘1.25e-01’ –
7	1	0.4591	0.7815	‘2.50e-01’, ‘9.88e-04’, ‘6.49e-03’, ‘9.88e-04’, ‘1.09e-02’, ‘6.49e-03’, ‘6.49e-03’, ‘9.88e-04’, ‘7.56e-02’, ‘9.88e-04’, ‘2.50e-01’, ‘6.49e-03’, ‘7.56e-02’, ‘7.56e-02’, ‘9.88e-04’, ‘7.14e-02’, ‘6.49e-03’, ‘7.14e-02’, ‘6.49e-03’, ‘7.56e-02’ –
8	0	0.3693	0.8528	‘1.48e-01’, ‘5.84e-04’, ‘3.84e-03’, ‘5.84e-04’, ‘3.54e-02’, ‘3.84e-03’, ‘3.84e-03’, ‘5.84e-04’, ‘4.47e-02’, ‘5.84e-04’, ‘1.48e-01’, ‘3.84e-03’, ‘4.47e-02’, ‘4.47e-02’, ‘5.84e-04’, ‘2.32e-01’, ‘3.84e-03’, ‘2.32e-01’, ‘3.84e-03’, ‘4.47e-02’ –
9	1	-0.2300	0.7628	‘8.99e-02’, ‘3.55e-04’, ‘2.34e-03’, ‘3.55e-04’, ‘2.15e-02’, ‘2.34e-03’, ‘2.34e-03’, ‘3.55e-04’, ‘1.25e-01’, ‘3.55e-04’, ‘8.99e-02’, ‘2.34e-03’, ‘1.25e-01’, ‘1.25e-01’, ‘3.55e-04’, ‘1.41e-01’, ‘2.34e-03’, ‘1.41e-01’, ‘2.34e-03’, ‘1.25e-01’ –
0	1	0.4591	0.7587	‘2.50e-01’, ‘2.17e-04’, ‘1.42e-03’, ‘2.17e-04’, ‘1.31e-02’, ‘1.42e-03’, ‘1.42e-03’, ‘2.17e-04’, ‘7.62e-02’, ‘2.17e-04’, ‘2.50e-01’, ‘1.42e-03’, ‘7.62e-02’, ‘7.62e-02’, ‘2.17e-04’, ‘8.62e-02’, ‘1.42e-03’, ‘8.62e-02’, ‘1.42e-03’, ‘7.62e-02’ –

9

Sample	x_1	x_2	y
0	0.4967	-0.1383	1
1	0.6477	1.5230	1
2	-0.2342	-0.2341	0
3	1.5792	0.7674	1
4	-0.4695	0.5426	1
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6	0.2420	-1.9133	0
7	-1.7249	-0.5623	0
8	-1.0128	0.3142	0
9	-0.9080	-1.4123	0
10	1.4656	-0.2258	1
11	0.0675	-1.4247	0
12	-0.5444	0.1109	0
13	-1.1510	0.3757	0
14	-0.6006	-0.2917	0
15	-0.6017	1.8523	1
16	-0.0135	-1.0577	0
17	0.8225	-1.2208	0
18	0.2089	-1.9597	0
19	-1.3282	0.1969	0

b

The initial log odds of the dataset is -0.8473

c

Sample	Residual
0	0.7
1	0.7
2	-0.3
3	0.7
4	0.7
5	-0.3
6	-0.3
7	-0.3
8	-0.3
9	-0.3

Sample	Residual
10	0.7
11	-0.3
12	-0.3
13	-0.3
14	-0.3
15	0.7
16	-0.3
17	-0.3
18	-0.3
19	-0.3

d and f

γ	values
$\gamma_{2,1}$	-1.4286
$\gamma_{3,1}$	1.7460
$\gamma_{4,1}$	3.3333
$\gamma_{2,2}$	-1.3715
$\gamma_{3,2}$	1.4696
$\gamma_{4,2}$	2.6719
$\gamma_{2,3}$	-1.3239
$\gamma_{3,3}$	1.2641
$\gamma_{6,3}$	2.2799
$\gamma_{2,4}$	-1.2837
$\gamma_{3,4}$	1.1036
$\gamma_{4,4}$	2.0190
$\gamma_{2,5}$	0.9553
$\gamma_{4,5}$	1.8327
$\gamma_{2,6}$	-1.2268
$\gamma_{3,6}$	0.9617
$\gamma_{4,6}$	1.6932
$\gamma_{2,7}$	-1.2006
$\gamma_{5,7}$	-0.3790
$\gamma_{6,7}$	1.8174
$\gamma_{2,8}$	-0.8550
$\gamma_{5,8}$	1.4880
$\gamma_{2,9}$	-1.1634
$\gamma_{5,9}$	-0.3550
$\gamma_{6,9}$	1.6501

γ	values
$\gamma_{2,10}$	-1.1454
$\gamma_{5,10}$	-0.3242
$\gamma_{6,10}$	1.5602

e

For tree 1, there are 3 leaf nodes: [2,3,4]

Samples for $\gamma_{2,1} = -1.4286$:

- 2, 5

Samples for $\gamma_{3,1} = 1.7460$:

- 0, 10

Samples for $\gamma_{4,1} = 3.3333$:

- 1, 3

10

Var	Sepal Length	Sepal Width	Petal Length	Petal Width	Target
Sepal Length	1	-0.117570	0.871754	0.81794	0.782561
Sepal Width	-0.117570	1	-0.428440	-0.366126	-0.426658
Petal Length	0.871754	-0.428440	1	0.962865	0.949035
Petal Width	0.81794	-0.366126	0.962865	1	0.956547
Target	0.782561	-0.426658	0.949035	0.956547	1

From our correlation matrix we find that the two features with the largest magnitude correlation with the target variable are the Petal Width and Petal Length features.

When training the OLS linear regression model, the testing had a MAE of 0.1359

When transforming the features into a degree 2 polynomial feature space and training an OLS linear regression on these features, the testing had a MAE of 0.1322